Perpendicular Lines

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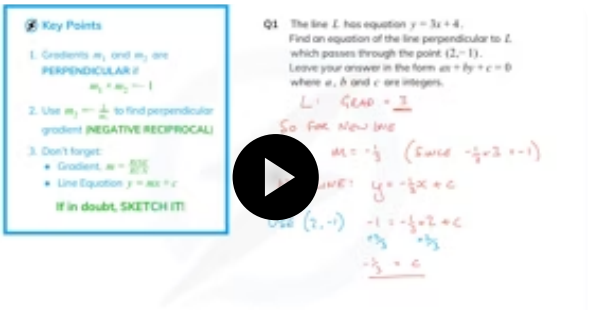

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Perpendicular lines

What are perpendicular lines?

- **Perpendicular lines** are **straight lines** which meet at **right-angles** (90°)
- One line may be referred to as a **normal** to the other line

How are the gradients of perpendicular lines related?

- Gradients m_1 and m_2 are **perpendicular** if $m_1 \times m_2 = -1$
  - For example
    - 1 and –1
    - $\frac{1}{3}$ and -3
    - $-\frac{2}{3}$ and $\frac{3}{2}$
- The two gradients are **negative reciprocals** of one another
- We can use $m_2 = -\frac{1}{m_1}$ to find a perpendicular gradient

How can I tell if two lines are perpendicular?

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

- $y = \frac{1}{3}x + 16$ and $y = -3x - 18$ **are perpendicular**
- $y = \frac{1}{7}x + 16$ and $y = 7x - 8$ **are not perpendicular**

- One or both of the equations may not be written in the form $y = mx + c$
  - In this case, you should **rearrange both equations into the form $y = mx + c$**
  - Their **gradients** can then be easily **compared**

How do I find the equation of a line perpendicular to another?

- You need to be able to find the **equation** of line that **passes through a particular point** and is **perpendicular** to another line
  - E.g. $5y = 4x + 30$ which **passes through** the **point** (8, 3)
- **Rearrange** the equation into the form $y = mx + c$ so that its gradient can be identified more easily
  - $y = \frac{4}{5}x + 6$
- Find the **gradient of the perpendicular** line
  - The gradient of the original line is $\frac{4}{5}$
  - Therefore the **gradient of the perpendicular** line is $-\frac{5}{4}$
  - The perpendicular line has an equation in the form $y = -\frac{5}{4}x + c$
- **Substitute the given point** into the equation for the perpendicular and **solve for** c
  - Substitute (8, 3), into $y = -\frac{5}{4}x + c$
  - $3 = -\frac{5}{4}(8) + c$
  - $c = 13$
- Substitute the value of c to find the **equation of the perpendicular**
  - The **equation** of the perpendicular line is $y = -\frac{5}{4}x + 13$
    - This could also be written as $4y = -5x + 52$ or equivalent

Worked Example

The line L has equation $y - 2x + 2 = 0$.

Find the equation of the line perpendicular to L which passes through the point $(2, -3)$.

Leave your answer in the form $ax + by + c = 0$ where a, b and c are integers.

gradient

$$y - 2x + 2 = 0$$
$$y = 2x - 2$$

Gradient of L is 2

The gradient of the line perpendicular to L will be the negative reciprocal of 2

$$m = -\frac{1}{2}$$

Substitute the point $(2, -3)$ into the equation $y = -\frac{1}{2}x + c$

Solve for c

$$y = -\frac{1}{2}x + c$$
$$-3 = -\frac{1}{2}(2) + c$$
$$-3 = -1 + c$$
$$c = -2$$

Write the full equation of the line

$$y = -\frac{1}{2}x - 2$$

The question asks for the line to be written in the form $ax + by + c = 0$ where a, b and c are integers

Move all the terms to the left hand side

$$\frac{1}{2}x + y + 2 = 0$$

Then multiply every term by 2, to ensure they are all integers

$$x + 2y + 4 = 0$$

How do I find the equation of a perpendicular bisector?

- A perpendicular bisector of a line segment cuts the line segment **in half** at a **right angle**
- Finding the **equation of the perpendicular bisector** of a line segment is very similar to finding the equation of a any perpendicular
  - Find the coordinates of the **midpoint** of the line segment
    - The perpendicular bisector will pass through this point
  - Find the **gradient of the line segment**
  - Then find the **negative reciprocal** of this gradient
    - This will be the **gradient of the perpendicular bisector**, m
  - Write the **equation** of the perpendicular bisector in the form $y = mx + c$

- Write the full **equation** of the perpendicular bisector in the form $y = mx + c$
- **Rearrange** the equation if the question requires a different form

Worked Example

Find the equation of the perpendicular bisector of the line segment joining the points (4, -6) and (8, 6).

Answer:

Find the coordinates of the **midpoint** of the line segment
The perpendicular bisector will pass through this point

$$\left(\frac{4+8}{2}, \frac{-6+6}{2}\right) = (6, 0)$$

Find the gradient of the line segment

$$\frac{-6-6}{4-8} = \frac{-12}{-4} = 3$$

Find the negative reciprocal of this
This will be the gradient of the perpendicular bisector, m

$$m = -\frac{1}{3}$$

Write the equation of the perpendicular bisector in the form $y = mx + c$

$$y = -\frac{1}{3}x + c$$

Substitute in the midpoint (6, 0) and solve to find c

$$0 = -\frac{1}{3}(6) + c$$
$$0 = -2 + c$$
$$c = 2$$

Write the full equation of the perpendicular bisector

$$y = -\frac{1}{3}x + 2$$

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